

GEMS TAIWAN
Elementary School Lesson Plan

Title/Subject : Plant Defenders: Beating Environmental Stress!

Target Group : Elementary school

Time : 1 hour 30 minutes

Course Objectives :

1. Understand the effects of global warming and its effects on Plant life (K)
2. Learn about elements of Synbio and how it can be used to solve real-world problems (K, A)
3. Get an understanding of photosynthesis and basic plant biology (K)
4. Learn about the type of effects abiotic and biotic factors have a plants (K)
5. Develop a basic understanding of DNA and its role in synthetic biology and the world (K, A)
6. Share what they learned about helping plants with others.
7. Suggest ways to care for plants in everyday life.

Teaching Flow :

1. Hook/ Introduction (5 minutes)

- Kahoot Pre-test
- Ask students 1-2 essential questions that are related to our topic today to their attention and interest
- Allow for discussion and sharing of answers and ideas
- Introduce the theme and contents for today's lesson

2. Important Vocabulary (5 minutes)

- Introduce a few important vocabulary words that are related to today's lesson
- Show slides with the word and definition in English and Chinese
- Ask them to repeat the words in english
- Explain the meaning and significance of the word

3. Photosynthesis & Plant Biology (15 Minutes)

- Show a short video that talks about the very basics of photosynthesis
- Have a short quiz on the contents of the video
- Recap everything said in the video
- Move on to plant stress and the causes of plant stress

4. Synthetic Biology and DNA (10 minutes)

- Show a video on an introduction to synthetic biology
- Explain the uses of synthetic biology
- Introduce the concept of how DNA works (ATGC base pairing)
- Show a short comic that explains what our team is doing for this years project

5. Experiment (20 minutes)

- Conduct an experiment to visualize photosynthesis using disks of leaves in water
- Show how abiotic stress like heat can affect how a plant is able to survive
- Allow for discussion of experiment results
- Teacher explains the experiment and the results in detail

6. DNA Themed Card Game (20 minutes)

- Play ATCG matching card game (a card game based on the original game: joker)
- Rules: A–T, C–G pairs; most pairs win.
- Learning: Reinforce DNA base pairs.

7. Arts & Crafts (10 minutes)

- Make a self-watering plant bottle using an empty plastic bottle
- Allow for fun, engaging, hands-on activity with a craft to keep

8. Post-Education Survey

- Ask students to fill out a survey regarding how they felt about the lesson today
- Include questions asking about knowledge of today's contents, how they feel about our lesson and how it affected their perspective on certain things, and any suggestions they have.

(Back-up plan / Played if there is extra time)

9. Review Game / End (10 minutes)

- Play a game of kahoot with multiple choice questions
- Goal: Review all contents of today's lesson as well as ending with an exciting game
- Include small prizes for winners

End